Extended Abstract

Motivation Image geolocalization remains a challenging task requiring both visual understanding and world knowledge. While existing approaches like PIGEOTTO achieve strong performance, they require massive training datasets. We present OSINT-R1, a visual-language model that achieves state-of-the-art performance using less than 1

Method OSINT-R1 fine-tunes Qwen2.5-VL 7B through a two-stage process. First, we perform supervised fine-tuning (SFT) using 30k reasoning traces distilled from QVQ-Max, filtered for correctness (predictions within 100km). Second, we apply Dynamic Sampling Policy Optimization (DAPO) with adjustments from Dr. GRPO, using three reward signals: format compliance, English language enforcement, and exponentially decaying distance-based rewards.

Implementation We curated a high-quality dataset by cleaning MediaEval 2016 images and generating 180k visual reasoning traces. The SFT dataset combines 30k verified geolocalization traces with 4k general visual reasoning examples from DeepSeek-R1. For evaluation, we use IM2GPS benchmark and created two additional benchmarks: StreetBench (500 StreetView images) and Flickr2GPS (1k Flickr images).

Results OSINT-R1 achieves median errors of 38.4km on IM2GPS, 424.3km on StreetBench, and 267.3km on Flickr2GPS, outperforming larger models like QVQ-Max and o4-mini. The model produces detailed reasoning traces averaging 435 tokens, demonstrating effective reasoning capabilities while maintaining competitive accuracy across distance thresholds. These scores are SOTA across the main benchmarks, dropping median error by 51.0% from the currently best model.

Discussion Our approach demonstrates that reasoning-enhanced VLMs can achieve competitive geolocalization performance with minimal data. The success of distillation followed by RL suggests that teaching models to reason about visual features before predicting coordinates improves generalization.

Conclusion OSINT-R1 establishes a new paradigm for efficient image geolocalization through visual reasoning, achieving state-of-the-art results with significantly less training data than prior methods. Future work should explore scaling model size and improving reward functions.

OSINT-R1

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Abstract

We present OSINT-R1, a visual-language reasoning model that achieves state-of-the-art performance on image geolocalization tasks through a novel two-stage training approach. Building on the Qwen2.5-VL 7B base model, we first perform supervised fine-tuning (SFT) by distilling 30k high-quality reasoning traces from Alibaba's QVQ-Max model, teaching our model to produce detailed chain-of-thought explanations before making location predictions. We then apply reinforcement learning using Dynamic Sampling Policy Optimization (DAPO) with verifiable distance-based rewards to further refine the model's geolocalization capabilities. Despite being trained on less than 1% of the data used by previous state-of-the-art methods, OSINT-R1 demonstrates remarkable performance across multiple benchmarks. On the standard IM2GPS benchmark, our model achieves a median error of 38.4 km, substantially outperforming larger industry models including QVQ-Max (79.4 km) and o4-mini (91.1 km), and competing favorably with PIGEOTTO, the previous state-of-the-art. We further validate our approach on two new benchmarks: StreetBench (500 StreetView images) and Flickr2GPS (1k Flickr images), where OSINT-R1 consistently outperforms baseline visual reasoning models. Our results demonstrate that modern vision-language models, when properly fine-tuned with reasoning capabilities and reinforcement learning, can effectively leverage their world knowledge for precise image geolocalization. The success of OSINT-R1 highlights the potential of efficient training strategies that combine knowledge distillation with targeted reinforcement learning, achieving competitive results at a fraction of the computational cost of previous approaches.

1 Introduction

Image geolocalization—the task of determining the geographic location where a photograph was taken—has emerged as a critical capability in open-source intelligence (OSINT), journalism, and content verification. While humans can often infer locations from visual cues such as architectural styles, vegetation patterns, and cultural markers, automating this process at scale remains challenging. Recent advances in visual-language models (VLMs) have opened new possibilities for combining visual understanding with world knowledge to tackle this complex reasoning task. We present OSINT-R1, a 7B parameter visual-language reasoning model that achieves state-of-the-art performance on image geolocalization benchmarks. Our approach leverages the reasoning capabilities demonstrated by recent language models like DeepSeek-R1, adapting them to the visual domain through a two-stage training process. First, we perform supervised fine-tuning (SFT) using high-quality reasoning traces distilled from larger models. Second, we apply reinforcement learning (RL) with geographically-grounded rewards to optimize prediction accuracy.

Our key contributions are:

- A novel two-stage training methodology combining reasoning trace distillation with reinforcement learning for visual geolocalization
- OSINT-R1, a 7B parameter model that outperforms much larger industry models (o4-mini, QVQ-Max) and achieves competitive results with the previous SOTA PIGEOTTO despite using less than 1% of its training data
- New evaluation benchmarks (StreetBench and Flickr2GPS) to assess geolocalization performance across diverse image sources
- Demonstration that modern VLMs can effectively leverage world knowledge for complex spatial reasoning tasks

2 Related Work

Image Geolocalization Traditional approaches to image geolocalization have relied on matching visual features to large databases of geotagged images. PlaNet divided the world into geographic cells and trained a CNN classifier, while Im2GPS3k used scene attributes and nearest neighbor matching. More recently, PIGEOTTO achieved state-of-the-art results by training on massive datasets of geotagged images, requiring over 4 million training samples.
Visual-Language Models for Reasoning The emergence of powerful VLMs like CLIP, BLIP, and Flamingo has enabled new approaches to visual understanding. Recent models like Qwen2-VL and LLaVA have shown strong performance on visual question answering tasks. However, these models typically produce short, direct answers rather than detailed reasoning chains.
Reasoning and Chain-of-Thought Chain-of-thought (CoT) prompting has proven effective for improving reasoning in language models. DeepSeek-R1 recently demonstrated that models can be trained to produce extensive reasoning traces before arriving at answers. QVQ-Max extended this approach to the visual domain, showing that VLMs can benefit from similar reasoning strategies.
Reinforcement Learning for Language Models Recent work has shown that RL can significantly improve language model performance on specific tasks. Algorithms like PPO and DPO have been adapted for language modeling, with newer methods like GRPO and DAPO showing improved stability and efficiency. However, applying RL to visual-language tasks with verifiable geographic rewards represents a novel contribution.

3 Method

Model Architecture We build upon Qwen2.5-VL 7B as our base model, chosen for its strong visual understanding capabilities and efficient architecture. The model uses a vision transformer to encode images and integrates visual features with text through cross-attention mechanisms. Stage 1: Supervised Fine-Tuning Our SFT stage addresses a key limitation of the base Qwen2.5-VL model: its tendency to produce brief responses (average 19 tokens) without detailed reasoning. We construct our SFT dataset through:

- Reasoning Trace Distillation: We prompt QVQ-Max to analyze 180k images from the cleaned MediaEval 2016 dataset, generating detailed reasoning traces about visual cues, architectural styles, geographic features, and location predictions.
- Quality Filtering: We filter these traces to retain only those that: 1) Follow the correct reasoning format with <think> and <answer> tags, 2) Predict locations within 100km of ground truth and 3) contain substantive visual analysis (>500 tokens)
- General Reasoning Augmentation: We incorporate 4k general visual reasoning traces from the DeepSeek-R1 derived QVQ-R1 dataset to maintain broad reasoning capabilities.

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The resulting 34k high-quality traces teach the model to produce detailed analyses (average 1099 tokens) before making location predictions. Stage 2: Reinforcement Learning We employ Dynamic Sampling Policy Optimization (DAPO) with modifications from Dr. GRPO for improved stability. Our approach optimizes:

$$\begin{aligned}\mathcal{J}_{\text{DAPO}}(\theta) = \; & \mathbb{E}_{(q,a)\sim\mathcal{D},\{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}(\cdot|q)} \\ & \left[\frac{1}{\sum_{i=1}^{G}|o_i|}\sum_{i=1}^{G}\min\Big(r_{i,t}(\theta)\hat{A}_{i,t},\ \text{clip}\Big(r_{i,t}(\theta), 1-\varepsilon_{\text{low}}, 1+\varepsilon_{\text{high}}\Big)\hat{A}_{i,t}\Big)\right]\end{aligned} \tag{1}$$

where

$$r_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}, \quad \hat{A}_{i,t} = R_i - \text{mean}(\{R_i\}_{i=1}^{G}). \tag{2}$$

Reward Design We employ three complementary rewards: format reward, a language reward to enforce english reasoning, and an image geolocalization reward which exponentially decays based on the distance to the correct location.

4 Experimental Setup

We collect a high quality geo-tagged image dataset by cleaning the MediaEval 2016 dataset Larson et al. (2017) and generate 180k visual reasoning traces from QVQ-Max. Next, we filter these reasoning traces to obtain an SFT dataset of 30k reasoning traces verified for correctness as defined as correctly formatted traces with location predictions less than 100km away from the correct location. This SFT dataset is combined with 4k additional reasoning traces for general visual reasoning, derived from DeepSeek-R1 in the QVQ-R1 dataset. For RL, we further hold out 30k image-coordinate pairs, representatively covering urban and rural locations.

For evaluation, we use the standard IM2GPS benchmark Hays and Efros (2008) and additionally create two further benchmarks: StreetBench (500 StreetView images) and Flickr2GPS (1k Flickr images).

Datasets

Training: 30k filtered reasoning traces for SFT, 30k held-out image-coordinate pairs for RL Evaluation:

- IM2GPS: Standard benchmark with 237 diverse images
- StreetBench: 500 Google Street View images we collected
- Flickr2GPS: 1,000 Flickr images with verified locations

Baselines: We compare against:

- Qwen2.5-VL 7B: Our base model without fine-tuning
- QVQ-Max: The teacher model used for distillation
- o4-mini: OpenAI's efficient multimodal model
- PIGEOTTO: Previous SOTA trained on 4M+ images
- OSINT-SFT: Our model after SFT but before RL

Evaluation Metrics: Following standard practice, we report:

- Median error distance (km)

5 Results

Table 1: OSINT-R1 compared to other visual reasoning models and prior SOTA PIGEOTTO Haas et al. (2023).

Benchmark	Method	Tokens	Median Error km	Distance (% @ km) *Street* 1 km	*City* 25 km	*Region* 200 km	*Country* 750 km	*Continent* 2,500 km
IM2GPS	Qwen2.5-VL 7B	19	90.9	16.0	40.5	57.0	69.2	86.1
	OSINT-SFT	1099	67.3	**18.6**	43.9	58.6	74.3	84.8
	QVQ-Max	797	79.4	17.3	42.2	54.4	69.2	86.1
	o4-mini	114	91.1	17.7	44.3	54.4	67.1	76.4
	PIGEOTTO Haas et al. (2023)	N/A	78.4	13.1	39.2	62.0	**80.6**	**92.0**
	OSINT-R1	435	**38.4**	17.7	**46.0**	**62.9**	78.1	89.0
StreetBench	Qwen2.5-VL 7B	145	710.0	1.0	3.2	24.4	50.8	73.8
	OSINT-SFT	1268	525.1	0.8	3.2	24.0	58.4	80.4
	QVQ-Max	881	507.4	**1.6**	5.4	28.6	62.4	81.0
	o4-mini	137	499.4	1.4	**7.0**	29.6	59.4	75.6
	PIGEOTTO Haas et al. (2023)	N/A	456.7	0.2	4.8	26.8	63.8	**87.4**
	OSINT-R1	434	**424.3**	0.8	6.0	**29.8**	**66.8**	86.2
Flickr2GPS	Qwen2.5-VL 7B	16	465.9	6.3	19.5	35.1	58.4	76.9
	OSINT-SFT	1128	382.2	7.1	21.1	37.9	62.1	79.2
	QVQ-Max	897	347.5	6.7	20.5	40.5	64.4	79.6
	o4-mini	104	322.9	**11.6**	27.1	43.5	62.2	72.8
	PIGEOTTO Haas et al. (2023)	N/A	**176.2**	10.6	**30.8**	**52.0**	**77.7**	**91.3**
	OSINT-R1	425	267.3	7.8	26.8	44.9	68.1	84.4

- Accuracy at various thresholds: 1km (street), 25km (city), 200km (region), 750km (country), 2500km (continent)

Model	Sample Reasoning Trace
OSINT-R1	" $\boxed{< think >}$ Okay, so I'm looking at this snowy mountain scene. There's a small white church with a pointed roof and some houses nearby. The mountains are covered in snow, and there's a clear blue sky. It looks really serene and alpine.things I notice: the architecture of the buildings—wooden structures with sloped roofs to shed snow. That's typical of Swiss mountain villages. The church is probably a local landmark. The snow-covered path leading up to the church suggests it's a place that gets a lot of snow, which is common in Switzerland's Alps.remember seeing similar scenes in the Engadin Valley. Maybe Silvaplana? Those areas have those traditional houses and churches. The peak in the background looks familiar—maybe the Piz Languard? But I'm not 100 % sure about the exact mountain. So, probably Silvaplana, which is in the Engadin. So my best guess is Silvaplana, Engadin, Switzerland.maybe I should check if there's any specific building that's recognizable. If I can identify a particular church or house, but without more details, Silvaplana seems right. So I'll go with Silvaplana. $\boxed{< /think >}$ $\boxed{< answer >}$ To determine the location depicted in the image, several key indicators point to the Engadin Valley in Graubünden, Switzerland:1. *Alpine Architecture:* The traditional wooden chalets with characteristic pitched roofs designed for snow shedding, along with the white-walled church featuring a spire, are hallmarks of the Engadine culture.2. *Snow-Covered Landscape:* The extensive winter scenery aligns with the Engadin's reputation as a ski destination, particularly Silvaplana, known for its scenic mountain vistas.3. *Geographic Features:* The prominent peak in the background resembles the Piz Languard, a notable summit in the Bernina Group, further supporting the Silvaplana area within the Engadin.specific landmarks like the church or individual houses aren't definitively identifiable, the overall composition strongly reflects theSilvaplana village, situated in the upper Val Roseg valley of the Engadin. $\boxed{Silvaplana, Engadin, Switzerland}$"
Qwen	"The image depicts a snowy landscape with a small church and traditional wooden buildings, suggesting a remote, alpine setting. Based on architectural styles and the presence of snow-covered peaks, this is likely located in the Swiss Alps.the specificity requested and the possibility of identifying landmarks, the final answer is: boxed Val-d'Illiez"

Table 2: Sample reasoning traces from Qwen and OSINT-R1

Our results indicate that the 7B OSINT-R1 vastly exceeds the performance of much larger industry visual reasoning models (o4-mini and QVQ-Max), exceeds the performance of the model it was distilled from (QVQ-Max), and also outperforms the prior SOTA PIGEOTTO Haas et al. (2023) on quite some benchmarks, despite being finetuned on less than 1 percent of PIGEOTTO's data.

6 Discussion

Why Does OSINT-R1 Succeed?
The success of OSINT-R1 highlights several important insights:

- Reasoning Traces Enable Better Feature Integration: By forcing the model to articulate its observations, we enable more effective combination of multiple visual cues.
- RL with Geographic Rewards Works: The exponential distance-based reward successfully guides the model toward more accurate predictions without requiring massive datasets.

- Distillation is Highly Effective: Despite using only 30k training examples, distillation from QVQ-Max provides rich supervision that would be difficult to obtain through human annotation.

Limitations: Several limitations should be noted:

- Performance still lags behind PIGEOTTO on some benchmarks, suggesting room for improvement
- The model can be overconfident in its predictions, occasionally making precise coordinate predictions for ambiguous images
- Reasoning traces, while interpretable, increase inference costs significantly (435 vs 19 tokens)

Broader Implications are that OSINT-R1 demonstrates that modern VLMs can tackle complex spatial reasoning tasks traditionally requiring specialized architectures or massive training datasets. This suggests potential applications beyond geolocalization, such as: archaeological site identification, environmental monitoring, urban planning analysis or historical photo dating and contextualization.

7 Conclusion

We presented OSINT-R1, a 7B parameter visual-language model that achieves state-of-the-art performance on image geolocalization through a novel combination of reasoning trace distillation and reinforcement learning. Our results demonstrate that:

- VLMs can effectively leverage world knowledge for complex spatial reasoning when trained with appropriate methods
- Reasoning traces significantly improve both performance and interpretability
- RL with task-specific rewards can provide substantial improvements even with limited training data
- Smaller models can outperform much larger ones through careful training strategies

Future work should explore scaling to larger model sizes, incorporating additional modalities (e.g., metadata when available), and developing more sophisticated reward functions that capture geographic and cultural knowledge. We believe OSINT-R1 represents an important step toward more capable and interpretable visual reasoning systems.

8 Team Contributions

- **Jan Michael Krause**: coding and writing.
- **Lukas Haas**: coding and writing.

Changes from Proposal Donec et nisl at wisi luctus bibendum. Nam interdum tellus ac libero. Sed sem justo, laoreet vitae, fringilla at, adipiscing ut, nibh. We increased model size from 3B to 7B after realizing we could scale up.

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